

ACME: Adaptive Cognitive Memory Engine

Externalizing Belief, Memory, and Learning from LLM Weights

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Code: <https://github.com/KamilBourouiba/ACME> (tag v0.1.1-arxiv)

Project site: <https://kamilbourouiba.github.io/ACME/>

Data: Production benchmark exports via GET /api/v1/benchmark/export (persisted in Postgres benchmark_runs)

Abstract

Large language models lack durable episodic memory, explicit belief states, structured forgetting, and mechanisms to learn from failure [1,2]. We present **ACME** (Adaptive Cognitive Memory Engine), an external cognitive substrate that treats the LLM as a language processor while delegating persistence, confidence tracking, contradiction handling, and self-correction to specialized engines. ACME operationalizes: *when does a collection of experiences deserve to become a belief?* We contribute (1) a promotion/demotion lifecycle with Cognitive Reliability Score (CRS), inspired by evidence-accumulation models in cognitive architecture [3,4]; (2) hybrid graph + pgvector retrieval [5,6] with contrarian verification [7,8]; (3) **MemoryBench v3** — thirteen sandbox-isolated scenarios with RAG [9], MemGPT [10], and LangGraph-style [11] baselines scored by an LLM judge [12,13]. On Azure OpenAI GPT-4.1 with `text-embedding-3-small` (256D) and Postgres Flexible + pgvector, ACME achieves overall **0.925** vs **0.487** for vector-RAG (MemoryBench v3, 24 June 2026 production run, job 3b31e5e3), with full feedback and belief-quality metrics unavailable to baselines. Model-sensitivity runs on GPT-4.1-mini and GPT-5.4 preserve a **+0.39–0.44** margin over RAG.

1. Introduction

Retrieval-augmented generation (RAG) [9] augments LLMs with document search but does not maintain explicit epistemic states, handle contradictory evidence, or learn from prediction failures. Dense and hybrid retrievers [14,15] improve recall yet remain stateless: they neither promote stable beliefs nor demote refuted hypotheses. Agent memory frameworks — MemGPT [10], generative agents [16], and LangGraph-style orchestration [11] — extend context windows via paging or state graphs but lack auditable belief lifecycles and structured feedback loops comparable to cognitive architectures [3,17].

Episodic and semantic memory have long been distinguished in psychology and AI [18,19]. Recent work on long-term conversational memory (MemGPT, memory streams [16], Zep [20]) focuses on *what to store and retrieve*, not *when accumulated evidence warrants belief*. ACME closes this gap with an explicit promotion pipeline (Observation -> Inference -> Hypothesis -> Belief) and symmetric demotion under contradiction — aligning with truth-maintenance and belief-revision traditions [21,22].

ACME externalizes cognition into specialized engines:

- **Episodic store** (PostgreSQL + pgvector [23]) and **semantic graph** (Neo4j [24], tenant-scoped)
- **Belief engine** — observation -> hypothesis -> belief -> challenged -> deprecated -> archived
- **Contrarian verification** on high-confidence answers [7,8]
- **Compression, forgetting, and autonomous learning** with prediction validation [25,26]
- **MemoryBench v3** for reproducible evaluation with CI gates

2. Related Work

Retrieval-augmented generation. Lewis et al. [9] established the retrieve-then-generate paradigm for knowledge-intensive tasks. Subsequent systems add re-ranking [14], query rewriting [27], and self-critique — Self-RAG [28] and Corrective RAG [29] route or revise retrieval based on generation quality. ACME differs by persisting structured beliefs and failure traces outside the LLM, not only refining a single-turn retrieval pass.

Graph-augmented memory. Knowledge-graph RAG [30,31] and GraphRAG [32] extract communities and summaries from document graphs. ACME maintains a *live* typed graph (entities, causal edges, tenant scope) updated on every experience, coupled with vector recall over episodic embeddings — hybrid retrieval akin to [5,6] but with belief-weighted context assembly.

Agent memory systems. MemGPT [10] virtualizes context via OS-inspired paging; generative agents [16] use memory streams and reflection; LangGraph [11] composes stateful agent workflows. These systems improve *capacity* and *session continuity*; ACME adds CRS-governed belief promotion, contradiction logging, and prediction-outcome loops absent from reference baselines in our evaluation.

Learning from feedback. Reflexion [33] and ExpeL [34] use verbal reinforcement from trial outcomes; Constitutional AI [35] shapes behavior via principles. ACME’s feedback engine ties outcome signals to specific graph-backed beliefs, adjusting confidence and status (e.g., Challenged $\square$ Deprecated) rather than only updating prompt-level reflections.

Cognitive architectures. ACT-R [3] and SOAR [17] model declarative memory, production rules, and learning from impasse. ACME is not a full cognitive architecture but borrows the separation of *fast reasoning* (LLM) from *slow accumulative memory* (Postgres, Neo4j, belief records) — similar in spirit to dual-process accounts [4] and complementary learning systems [36].

Evaluation of memory systems. Benchmarks such as LongMemEval [37], LoCoMo [38], and MemoryBank [39] stress long-horizon recall. MemoryBench v3 additionally scores feedback correction, belief quality, hallucination resistance under adversarial queries, and per-scenario sandbox isolation — metrics orthogonal to pure retention [12,13].

3. System Design

Ingestion: LLM extraction [40] plus deterministic normalization -> Neo4j entities/relations + episodic embeddings [23].

Query: Hybrid retrieval (graph neighborhood [24,30] + pgvector cosine [5]) -> LLM reasoning -> optional contrarian pass [7,8] when confidence ≥ 0.8 .

Feedback: Outcome signals update beliefs [21], log failures [33], trigger consolidation (scheduled consolidation, 6 h).

CRS (Cognitive Reliability Score) = 40% prediction success + 20% temporal stability + 20% contradiction resistance + 20% source diversity — a composite reliability index analogous to calibrated confidence under heterogeneous evidence [41,42].

Causal edge types: `observed_with`, `precedes`, `correlates`, `causes`, `disproves` — distinguishing correlation from causation per Pearlian and epistemic traditions [43,44].

Multi-tenancy: Postgres rows and Neo4j nodes keyed by `tenant_id` (header X-Tenant-ID).

Ablation toggles: `ABLATION_DISABLE_CONTRARIAN`, `ABLATION_DISABLE_BELIEF_SYNC`, `ABLATION_DISABLE_VECTOR` (see Section 6).

4. MemoryBench v3

Four metrics, LLM-as-judge [12,13] (semantic; keyword overlap reported separately):

Metric	Description
Retention	Concept coverage (synonyms accepted)
Groundedness	Answer supported by ingested episodes [45,46]
Feedback correction	Belief adjustment after contradictions [21,33]
Belief quality	Mean CRS across tracked beliefs

Overall = average of all four. Baselines score 0 on feedback/belief (N/A), capping baseline overall near 0.48–0.49.

Thirteen scenarios: retention, contradiction, multi-source conflict, error injection, long-term retention, hallucination resistance, feedback adjustment, adversarial noise, long-horizon noise, tenant isolation, healthcare domain transfer, multi-session recall, prediction-outcome loop.

Isolation: Each scenario deletes prior benchmark-tagged Postgres rows and Neo4j subgraph before run (`acme/evaluation/sandbox.py`) — preventing cross-scenario contamination noted as a failure mode in long-horizon evals [37,38].

Baselines: Minimal reproducible implementations aligned with Lewis et al. [9] (RAG), Packer et al. [10] (MemGPT), and LangGraph-style state graphs [11] — see `docs/BASELINES.md`.

5. Experimental Setup

Component	Configuration
LLM	Azure OpenAI GPT-4.1 [47]
Embeddings	<code>text-embedding-3-small</code> , 256 dimensions [48]
Database	Azure Postgres Flexible (pgvector [23]), francecentral
Graph	Neo4j 5.x on Azure Container Apps [24]
Judge	Same deployment as extractor/reasoner (<code>evaluate_answer_quality</code>) [12]
Reproducibility	<code>POST /api/v1/benchmark/memorybench</code> , <code>POST /api/v1/benchmark/compare/async</code>

Model-sensitivity runs additionally use GPT-4.1-mini and GPT-5.4 (same endpoint family; GPT-5.4 requires `max_completion_tokens` API parameter).

6. Results (Azure production, 24 June 2026, 13 scenarios, isolated)

Run metadata: API revision `acme-api--membenchfix`; compare job `3b31e5e3-72b1-4ce3-b6c2-848cf94e4128`; duration 725 s; zero scenario failures.

System	Retention	Groundedness	Feedback	Belief Q.	Overall
ACME	1.000	1.000	1.000	0.700	0.925
RAG baseline	0.969	0.977	N/A	N/A	0.487
MemGPT baseline	0.977	0.969	N/A	N/A	0.487

System	Retention	Groundedness	Feedback	Belief Q.	Overall
LangGraph baseline	0.900	0.977	N/A	N/A	0.469

Baselines exclude feedback/belief in overall (not applicable). Scores rounded to three decimals from persisted `benchmark_runs` export.

Key finding: ACME gains **+0.44** overall vs RAG primarily from feedback correction and belief quality — dimensions absent from all baselines [10,11]. Retention and groundedness are competitive across systems (all ≥ 0.90 on applicable metrics), consistent with strong vector and graph-augmented recall [9,32].

Model sensitivity (GPT-4.1-mini): Job 107d02c0, 639 s. ACME retains **+0.39** overall vs RAG (0.858 vs 0.473). Feedback stays at 1.000 and belief at 0.700 — cognitive layers compensate when the base model is weaker [4,36].

System	Retention	Groundedness	Feedback	Belief Q.	Overall
ACME	0.892	0.838	1.000	0.700	0.858
RAG baseline	0.908	0.985	N/A	N/A	0.473
MemGPT baseline	0.915	1.000	N/A	N/A	0.479
LangGraph baseline	0.854	0.862	N/A	N/A	0.429

Model sensitivity (GPT-5.4): Job eb68f028, 620 s. ACME **0.873** vs RAG **0.474 (+0.40)**.

System	Retention	Groundedness	Feedback	Belief Q.	Overall
ACME	0.952	0.839	1.000	0.701	0.873
RAG baseline	0.964	0.931	N/A	N/A	0.474
MemGPT baseline	0.975	0.939	N/A	N/A	0.478
LangGraph baseline	0.973	0.950	N/A	N/A	0.481

Belief quality remains **~0.70** across all three LLM tiers — suggesting the external belief substrate [3,21] is comparatively stable while retention/groundedness track model capability [12].

Ablation (design): Disabling contrarian checks, belief sync, or vector retrieval is supported via environment flags; unit gates verify toggles (`scripts/run_ablation_gate.py`). Full ablation sweeps on live LLM runs are left to future work.

7. Limitations

- LLM judge and extractor introduce variance [12,13]; we report sandbox-isolated runs with persisted `benchmark_runs`.
- Scenarios emphasize SaaS churn/latency plus one healthcare transfer set; broader domains and multilingual evals needed [37,38].
- Baselines are reference implementations, not full upstream MemGPT/LangGraph codebases [10,11].
- Production API requires API key for benchmark endpoints.
- CRS weights are hand-tuned; calibration against human epistemic judgments [41] is future work.

8. Conclusion

ACME shows that externalizing belief and learning from LLM weights yields stronger cognitive memory than vector RAG or lightweight agent-memory baselines on MemoryBench v3, especially for feedback-driven correction and auditable belief quality. The consistent **+0.39–0.44** margin over RAG across GPT-4.1, GPT-4.1-mini, and GPT-5.4 supports the hypothesis that explicit belief lifecycle management [21,22] complements — rather than replaces — advances in base model capability [1,47].

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Appendix A — Reproduction

```
git clone https://github.com/KamilBourouiba/ACME.git && cd ACME
pip install -e ".[dev]"
pytest tests/test_benchmark_gate.py tests/test_memorybench.py -q
# Production (requires API key):
./azure/configure-premium-ingress.sh
./scripts/run_prod_benchmark.sh
./scripts/run_model_benchmark.sh gpt-4.1-mini # optional model sweep
```

Appendix B — Data Availability

Benchmark payloads stored in PostgreSQL table `benchmark_runs` (columns: `overall_score`, `retention_score`, `belief_quality_score`, `payload JSONB`, `created_at`). Export via `GET /api/v1/benchmark/export`. Full run tables: `docs/BENCHMARK_RESULTS.md` [49].